

A

Project Synopsis

On

Diabetes Prediction

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1. INTRODUCTION

1.1 PROBLEM BACKGROUND

Diabetes mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels due to insulin deficiency or resistance. According to the World Health Organization (WHO), diabetes is one of the leading causes of morbidity worldwide and contributes significantly to cardiovascular diseases, kidney failure, and premature mortality.

Early detection is critical to reduce long-term complications. Traditional diagnosis relies on laboratory testing and clinical evaluation. However, manual risk assessment:

- May overlook complex feature interactions
- Is time-consuming
- Lacks predictive modelling support

Machine Learning (ML) provides automated, data-driven risk assessment systems capable of identifying high-risk patients early.

1.2 RELATED WORK :

Current research on diabetes prediction has progressed from traditional statistical approaches such as logistic regression applied to datasets like the Pima Indians Diabetes Dataset to advanced machine learning classifiers including Support Vector Machine and Random Forest for binary (0/1) prediction. Recent studies emphasize ensemble boosting techniques such as XGBoost to enhance predictive accuracy. Researchers also incorporate preprocessing strategies like SMOTE for class imbalance and MICE for missing data handling. Furthermore, explainable AI methods such as SHAP and clinical risk scoring tools like FINDRISC are being explored to improve interpretability and risk assessment.

Research Papers studied :

Approach	Core Contribution	Limitation
Boosting-Based Models [1], [2]	High predictive accuracy for diabetes classification	Lack interpretability and clinical decision support
Imbalance Handling (SMOTE) [3]	Improves minority class detection	Does not address noise and full pipeline optimization
Feature Selection (Boruta) [4]	Identifies statistically significant predictors	Not integrated into complete ML framework
Missing Data Imputation (MICE) [5]	Handles incomplete clinical data effectively	Rarely combined with optimized ensemble learning
Explainable AI (SHAP) [6]	Provides feature-level prediction explanation	No hybrid diabetes scoring mechanism
Clinical Risk Score (FINDRISC) [7]	Lifestyle-based diabetes risk estimation	Static, non-adaptive, lower predictive power

1.3 RESEARCH GAPS IDENTIFIED :

1. **Absence of an Integrated End-to-End Framework**

Existing studies address preprocessing, imbalance handling, feature selection, modeling, or scoring independently, but lack a unified and optimized predictive pipeline.

2. **Limited Clinical Interpretability**

High-performing machine learning models do not adequately provide transparent, patient-specific explanations suitable for clinical decision-making.

3. **Inadequate Hybrid Risk Stratification**

Clinical lifestyle-based scoring systems and machine learning probability outputs are not combined into a comprehensive diabetes meter or unified risk score.

4. **Suboptimal Data Handling Strategies**

Missing data imputation, imbalance correction, and feature validation are rarely implemented together in a statistically rigorous manner.

5. **Lack of Personalized Decision Support**

Current approaches focus primarily on classification (0/1 outcome) without integrating actionable lifestyle-based recommendations for risk reduction.

1.4 PROBLEM STATEMENT

To develop an **optimized, accurate, and explainable machine learning model** for early diabetes prediction by integrating:

- Advanced preprocessing
- Hybrid resampling
- Robust feature selection
- Boosting-based classifiers
- SHAP-based interpretability

1.5 OBJECTIVES :

- To preprocess medical data using MICE imputation and IQR-based outlier removal.
- To address class imbalance using SMOTE–Tomek hybrid resampling.
- To perform feature selection using Boruta algorithm.
- To develop a boosted tree model (XGBoost/CatBoost) with Bayesian hyperparameter optimization.
- To provide global and local explanations using SHAP.
- To deploy the model via an interactive web interface (Streamlit/Flask).

2. TECHNOLOGIES USED :

2.1 Dataset :

- PIMA Indian Diabetes Dataset
Source: [UCI Machine Learning Repository](#)

Features include:

- Glucose
- BMI
- Blood Pressure
- Insulin
- Age
- Pregnancies
- Diabetes Pedigree Function

Target:

- Outcome (0 = Non-diabetic, 1 = Diabetic)

2.2 Machine Learning Techniques

Preprocessing

- MICE (Iterative Imputer)
- IQR Outlier Removal

Balancing

- SMOTE-Tomek

Feature Selection

- Boruta

Modelling

- XGBoost / CatBoost
- Bayesian Optimization (Optuna)

Interpretation

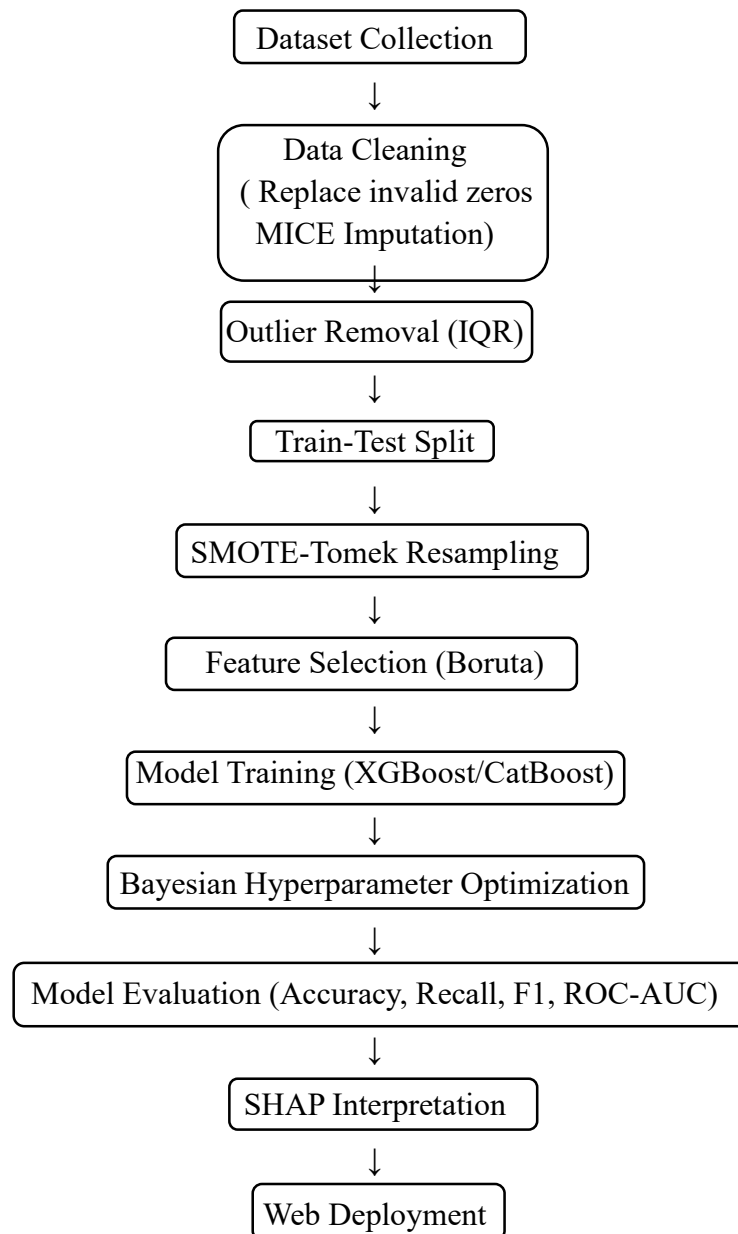
- SHAP (TreeExplainer)

2.3 Libraries Used

- NumPy
- Pandas
- Scikit-learn
- Imbalanced-learn
- XGBoost
- CatBoost
- Optuna
- SHAP
- Matplotlib / Seaborn
- Streamlit (for deployment)

3. PROPOSED METHODOLOGY

3.1 Workflow :



1. Dataset Collection :

The PIMA Indian Diabetes dataset is obtained from the UCI Machine Learning Repository.

It contains clinical diagnostic features such as glucose level, BMI, age, and insulin, with a binary target indicating diabetic or non-diabetic status.

2. Data Cleaning

Invalid zero values in medical attributes (e.g., Glucose, BMI) are replaced with missing indicators. MICE (Multiple Imputation by Chained Equations) is then applied to estimate missing values using regression-based iterative modelling.

3. Outlier Removal (IQR)

Outliers are detected using the Interquartile Range method: values outside $1.5 \times \text{IQR}$ are treated as extreme.

This reduces noise and prevents model bias caused by abnormal clinical measurements.

4. Train-Test Split

The dataset is divided into training and testing sets (e.g., 80:20 ratio).

Training data is used to build the model, while test data evaluates its generalization performance.

5. SMOTE–Tomek Resampling

SMOTE generates synthetic minority-class samples to balance the dataset.

Tomek Links remove borderline overlapping samples, improving class separability and recall.

6. Feature Selection (Boruta)

Boruta is a wrapper-based method built on Random Forest that selects only statistically significant features.

It eliminates irrelevant biomarkers and retains the most impactful predictors.

7. Model Training (XGBoost / CatBoost)

Gradient boosting algorithms such as XGBoost or CatBoost are trained on the selected features.

These models capture nonlinear relationships and feature interactions effectively.

8. Bayesian Hyperparameter Optimization

Optuna-based Bayesian optimization searches for optimal hyperparameters (e.g., learning rate, depth).

This improves model accuracy while preventing overfitting.

9. Model Evaluation

Performance is assessed using Accuracy, Recall, F1-score, and ROC-AUC.

Recall is particularly emphasized to minimize false negatives in medical prediction.

10. SHAP Interpretation

SHAP explains predictions by assigning contribution values to each feature.

It provides both global importance ranking and patient-specific explanations.

11. Web Deployment

The final trained model is integrated into a web interface (e.g., Streamlit).

Users can input patient data and receive prediction results with visual explanations.

3.2 Expected Outcomes :

- Improved recall (10–15%) compared to baseline.
- Accuracy expected in range: 95–98% (with tuning).
- Reduced training time due to feature selection.
- Transparent model with patient-specific medical justification.
- Deployable clinical decision-support prototype.

3.3 Conclusion :

This project proposes an optimized and explainable machine learning framework for diabetes prediction by integrating advanced preprocessing, hybrid resampling, wrapper-based feature selection, boosting classifiers, and SHAP interpretability.

Unlike traditional models that operate as black boxes, this framework emphasizes transparency and reliability, making it suitable for medical decision-support applications.

The final system aims to provide not only high predictive performance but also interpretable risk explanations for each patient, thereby increasing trust and usability in healthcare environments.

References :

1. **XGBoost: A Scalable Tree Boosting System** – Chen & Guestrin (2016)
2. **CatBoost: Unbiased Boosting with Categorical Features** – Dorogush et al. (2018)
3. **SMOTE: Synthetic Minority Over-sampling Technique** – Chawla et al. (2002)
4. **Feature Selection with the Boruta Package** – Kursa & Rudnicki (2010)
5. **Multivariate Imputation by Chained Equations (MICE)** – van Buuren & Groothuis-Oudshoorn (2011)
6. **A Unified Approach to Interpreting Model Predictions (SHAP)** – Lundberg & Lee (2017)
7. **Finnish Diabetes Risk Score (FINDRISC)** – Lindström & Tuomilehto (2003)